

LVMH Digital Product Passport

A project for the LVMH Hackathon at Albert School, aiming to transform the Digital Product Passport (DPP) from a regulatory requirement into a powerful tool for building trust, prestige, and long-term client engagement.

✦ Project Vision

Our vision is to transform the mandatory disclosure of the Digital Product Passport into a powerful lever for trust, prestige, and long-term client engagement for Louis Vuitton. We've created a comprehensive framework and strategic roadmap that integrates technical architecture with a premium user experience, turning regulatory mandates into brand opportunities.

🌀 Core Concept: The DPP Emotion Funnel

The user experience is designed around a four-stage emotional journey:

1. **Attraction:** Immersive product storytelling to capture emotional attention.
2. **Confidence:** A verifiable Certificate of Authenticity, anchored on the Aura Blockchain.
3. **Empowerment:** A lifelong care and longevity hub, with aftercare tutorials and service information.
4. **Serenity:** Transparent and compliant regulatory information.

This funnel is designed to seamlessly transition the client from subconscious desire to conscious trust.

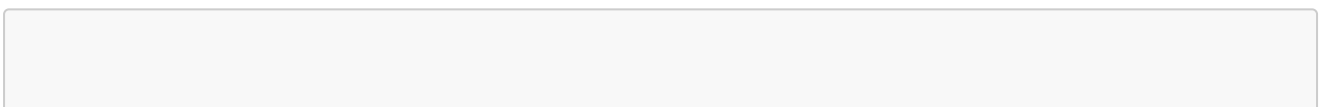
💡 The Key Differentiator: Invitation-Only Events

By linking a product's unique ID to a user's account, we can organize exclusive, invitation-only events for product owners. This reinforces the brand's prestige and drives customer lifetime value.

⬇️ User flow and backend orchestration

1. **Anonymous Visitor:** Lands on the site and is redirected to the [/collection](#) page. If not logged in, they are prompted to create an account.
2. **Account Creation:** A simple form (pre-filled for demo purposes) allows the user to create an account.
3. **Collection Page:** The user sees their collection of products. They can click on a product to view its DPP.
4. **Product Owner on DPP Page:** When a logged-in user views a DPP page, they see the full details, including the unblurred certificate number and access to the service ledger.
5. **Transfer of ownership:** A logged-in user can initiate the transfer of one of their authenticity certificates to another user. The backend system will carry out the ownership change, including the corresponding update on the blockchain.

The DPP framework follows a detailed flow, from the initial scan to the final ownership certification:



```

graph TD
    A[User scans<br>the NFC chip] --> B[That opens the<br>public product page]
    B --> C["The page shows<br>Storytelling, details<br>and a login button"]
    C --> D["User clicks on<br>the login button"]
    D --> E["User logs in or<br>creates an account"]
    A --> I[UID is stored<br>in cache]
    I --> F
    E --> F["Product UID & customer ID<br>are sent to backend"]
    F --> G[Backend links IDs,<br>creates Certificate ID &<br>writes on Aura Blockchain]
    G --> H[The certified card<br>appears in the<br>user collection]
    G --> J
    E --> H
    J --> F
    H --> J[The user transfer his card<br>to another user]

    %% -----
    %% Styles
    classDef public fill:#d1fae5,stroke:#059669,stroke-width:2.5px
    classDef user fill:#fef3c7,stroke:#d97706,stroke-width:2.5px
    classDef backend fill:#e9d5ff,stroke:#b95cdb,stroke-width:2.5px
    classDef newuser fill:#c3d1f7,stroke:#2c5de6,stroke-width:2.5px

    class B,C public
    class A,D,E,H user
    class F,G,I backend
    class J newuser

    %% -----
    %% user flow = orange / backend flow = violet
    linkStyle 0,1,2,3,10 stroke:#d97706,stroke-width:2.5px
    linkStyle 4,5,6,7,8,9 stroke:#b95cdb,stroke-width:2.5px
    linkStyle 11,12 stroke:#2c5de6,stroke-width:2.5px

```

```

graph TD
    P[Public steps]:::public
    U[User flow]:::user
    K[Backend flow]:::backend
    L[New user transfer]:::newuser

    %% Styles
    classDef public fill:#d1fae5,stroke:#059669,stroke-width:2.5px
    classDef user fill:#fef3c7,stroke:#d97706,stroke-width:2.5px
    classDef backend fill:#e9d5ff,stroke:#7c3aed,stroke-width:2.5px
    classDef newuser fill:#c3d1f7,stroke:#2c5de6,stroke-width:2.5px

```

The project is built on a modern, robust, and scalable technical stack:

- **Frontend:** Next.js 16, React 19, TypeScript
- **Styling:** TailwindCSS 4, Framer Motion for animations
- **UI Components:** Radix UI
- **Blockchain:** Aura Blockchain for product authenticity
- **Authentication:** A simplified demo using localStorage

Project Structure

The repository is organized as follows:

```
app/  
├─ page.tsx                # Redirects to /collection  
├─ products/[slug]/page.tsx # Dynamic DPP page for each product  
├─ account-creation/  
└─ collection/             # User's product collection dashboard  
  
components/  
├─ page-dpp-section-[1-4].tsx # The four sections of the DPP page  
├─ feature-certificate-card-v3.tsx # Interactive 3D certificate card  
├─ feature-collection-grid.tsx    # Grid of products in the user's  
collection  
└─ ...                           # Other reusable components  
  
contexts/  
└─ user-context.tsx           # Manages user state and authentication  
  
lib/  
└─ products.ts               # Product data and helper functions
```

App Architecture

The app is designed to be highly scalable and responsive, everything is "variable made" to allow any products to be added, we just have to drop images in the public/products folder and update the lib/products.ts with his informations.

Deployment

This project is designed to be deployed as a static site on **GitHub Pages**. No complex installation or build process is required. The repository can be cloned and the **out** directory can be served statically.

[---> ACCESS TO THE DEPLOYED WEBSITE HERE <---](#)